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ENGINEERING

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Sreevetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	T.PRAKEERTHI	Register Number	192524279
Class / Section		Slot	B
Course Code	CSA1512	Course Title	Cloud computing
Capstone Title	Medi Queue AI: Hospital of Queue and No-show. prediction App.		
Submission Date	13 July 2026		
Faculty Name	Dr. Rajesh Babu	Employee ID	
Department			

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	
Date	

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

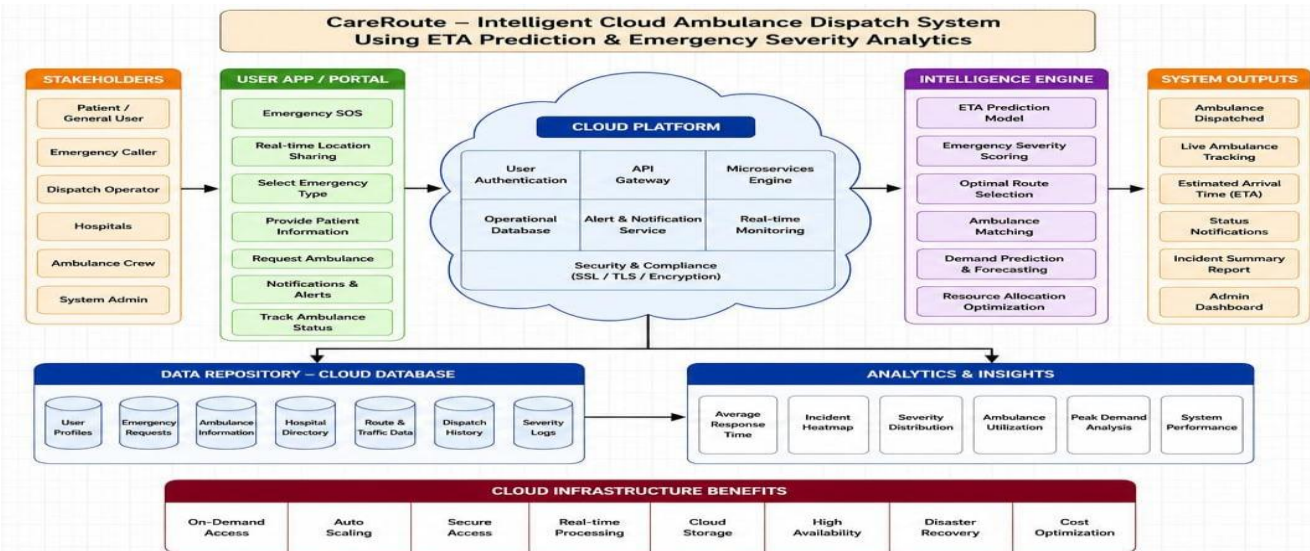
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

MediQueue AI is an AI-powered hospital OP (Outpatient) Queue and No-Show Prediction application designed to improve patient flow and reduce waiting time. The concept map is centered on the **MediQueue AI** node, which branches into four major modules: **Patient Module**, **AI Prediction Module**, **Hospital Management Module**, and **Cloud Database**. The Patient Module allows users to book appointments, view queue status, and receive reminders. The AI Prediction Module analyzes patient history and appointment data to predict possible no-shows, while the Hospital Management

Module helps doctors and staff monitor appointments and manage queues efficiently. The Cloud Database securely stores patient records, appointment details, and prediction results.

The link labels in the concept map describe how the modules interact, using connections such as **books**, **predicts**, **stores**, **sends**, and **monitors**. The hierarchy starts with the main concept, followed by the major modules, and finally the individual features like appointment booking, queue tracking, notifications, dashboard, and prediction model. The capstone connection highlights the integration of **Artificial Intelligence**, **Cloud Computing**, and **Healthcare Management** to solve real-world hospital challenges. By predicting patient no-shows and optimizing OP queues, MediQueue AI improves resource utilization, reduces waiting times, and enhances the overall patient experience.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS				
PaaS				
SaaS				
XaaS / Specialized Service				

Task 2B - Capstone Cloud Service Decision Matrix

Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI				
API / backend logic				
Authentication and identity				
Database / structured records				
File / object storage				
Monitoring / logs / usage metrics				
Analytics / reporting output				

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

Technical Justification

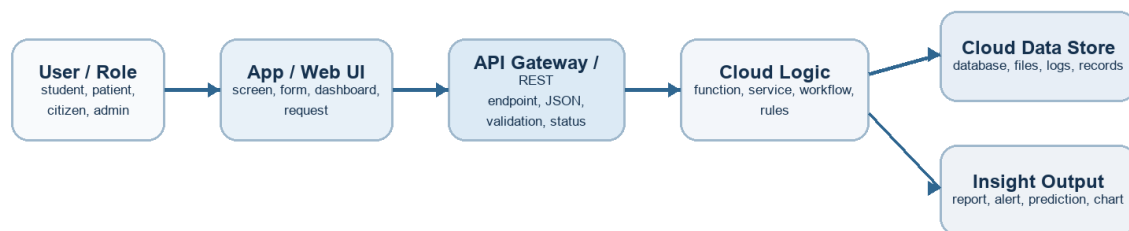
MediQueue AI is technically justified because it combines Artificial Intelligence (AI), Cloud Computing, and real-time data processing to solve hospital OP queue management problems. The AI model analyzes patient appointment history, attendance patterns, and other relevant data to predict possible no-shows, enabling hospitals to optimize appointment scheduling and reduce idle time. Cloud services provide secure storage, scalability, and real-time access to patient and appointment data, while notification services automatically remind patients of upcoming visits. This architecture improves system efficiency, reduces patient waiting time, increases doctor utilization, and provides a reliable, scalable, and secure solution for modern healthcare management.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

Sample Capstone Cloud Flow Template

Replicate this structure using your own capstone screens, APIs, cloud services, data stores, and outputs.



Cross-cutting cloud decisions to label in your own diagram

Authentication / IAM

Who is allowed to access each API and data item?

Scalability

Which component must handle growing users or records?

Elasticity

Which component can expand or shrink based on demand?

Monitoring

What logs, usage metrics, or errors will be observed?

Security

Where are validation, privacy, backup, and access control applied?

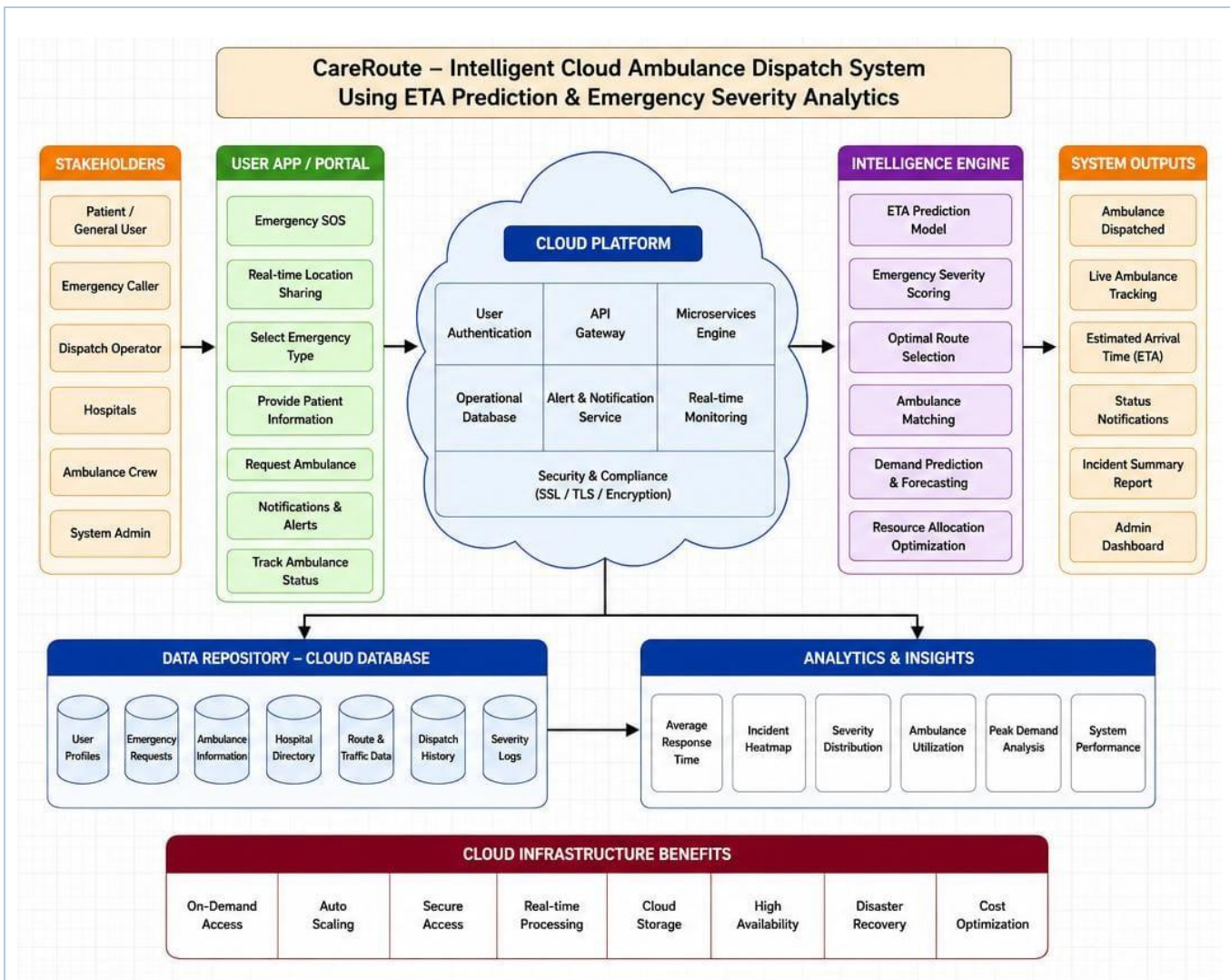
Student instruction: redraw this flow in draw.io/diagrams.net and replace every label with your own capstone architecture.

Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	
Frontend / app screen	
API endpoint and method	
Backend processing rule	
Cloud database / storage operation	
Response, alert, report, or dashboard output	
Security, scalability, and elasticity labels added in diagram	

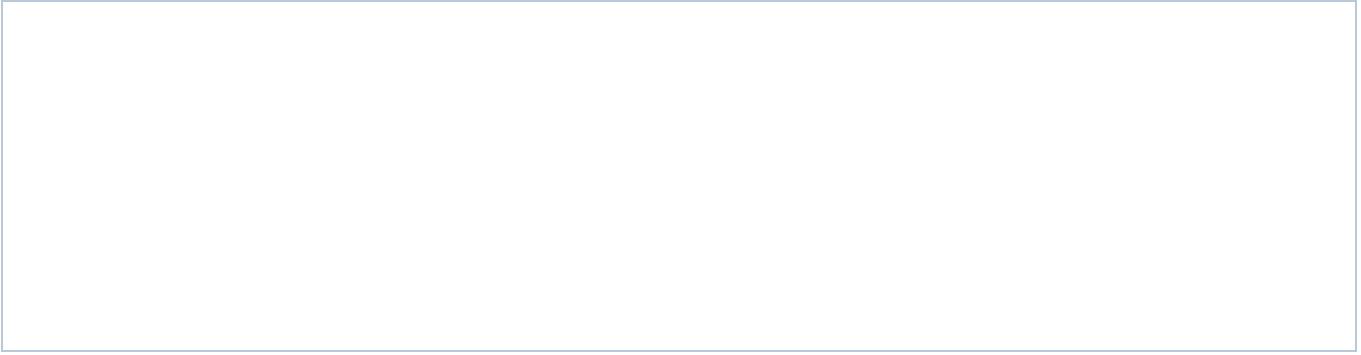
Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

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A large, empty rectangular box with a thin blue border, intended for the student's individual submission. It occupies the upper half of the page.

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud concept in my capstone project is **cloud-based data storage and real-time processing**. I think it is important because the application stores attendance records securely and provides instant updates and early-warning alerts.

2. Where will elasticity or scalability become important in your capstone product?

Elasticity and scalability become important when many students and faculty use the application at the same time. The cloud can automatically increase resources to maintain good performance without slowing down the system

3. Which API or web-service idea is most relevant to your project and why?

The most relevant web-service idea is the **REST API**. I understood that it helps the mobile app, web application, and cloud database communicate with each other quickly and securely for real-time attendance updates

4. What did you personally understand better after preparing this concept map and matrix? After preparing the concept map and matrix, I understood how different cloud computing concepts such as service models, deployment models, APIs, scalability, and security are connected and how they work together in a real-world application like my capstone project.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.

Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or generic non-technical content.
Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	